HUNDRED DAYS OF CODE
DAY 12

QUESTION : Write a menu-driven C program using switch-case to input a positive integer and find the sum and product of the first and last digits. **E.g.** Number: 12345, Sum = 6, Product = 5.

INPUT: A positive integer n

Menu choice 1

OUTPUT: The program displays the sum and product of the first and last digits.

ALGORITHM:

1. Start.
2. Input a positive integer n.
3. Display the menu.
4. Input the choice.
5. Use switch-case.
6. For choice 1:

- Store n in a temporary variable first.
- Find the last digit using n % 10.
- Find the first digit by repeatedly dividing first by 10 until it becomes a single digit.
- Calculate sum = first + last.
- Calculate product = first × last.
- Display the sum and product.

7. Stop.

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PSEUDUCODE:

```
START

INPUT n
INPUT choice

SWITCH choice

CASE 1:

  first ← n
  last ← n MOD 10


  WHILE first >= 10 DO
    first ← first DIV 10

  END WHILE


  sum ← first + last
  product ← first × last


  OUTPUT sum
  OUTPUT product


DEFAULT:
  OUTPUT "Invalid choice"


END SWITCH


STOP
```

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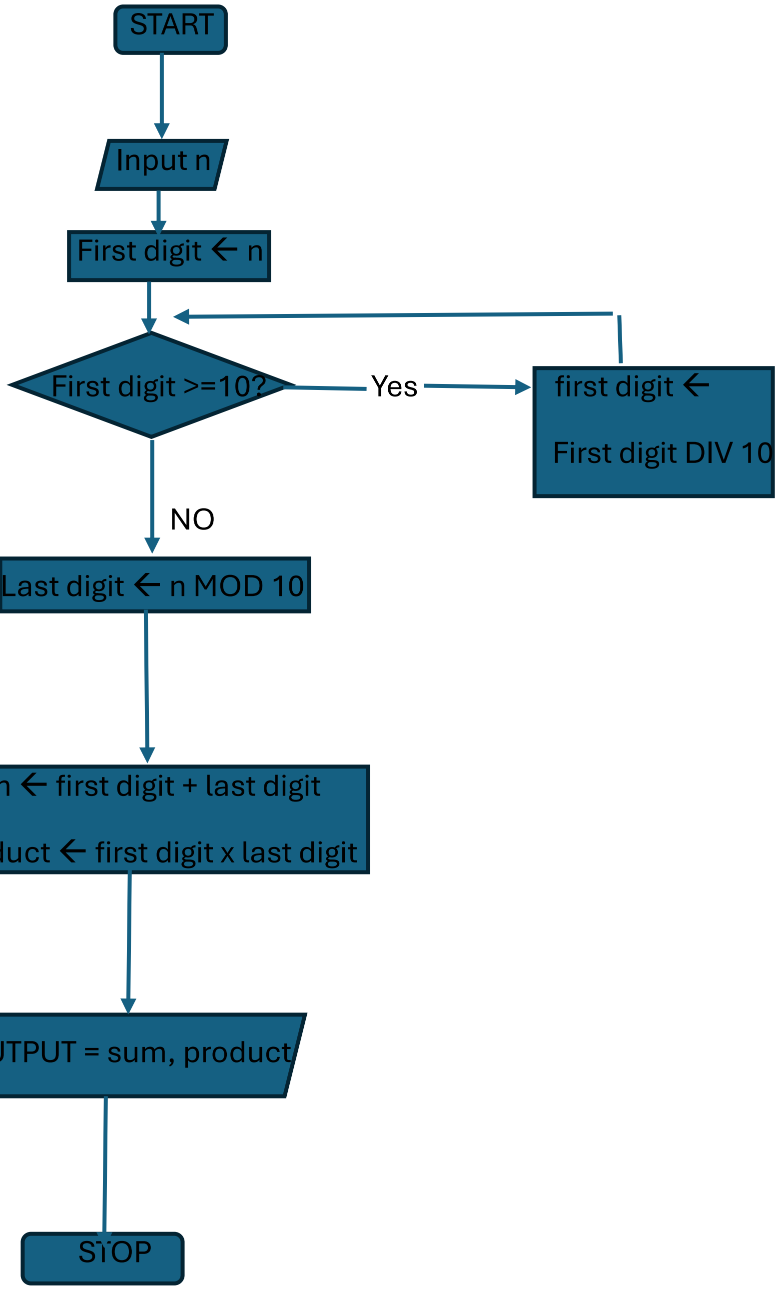
FLOWCHART:



TEST CASE TABLE:

TEST CASE	CHOICE	INPUT	EXPECTED OUTPUT	ACTUAL OUTPUT	RESULT
1	1	12345	SUM=6, PRODUCT=5	SUM=6, PRODUCT=5	PASS
2	1	5	SUM=10, PRODUCT=25	SUM=10, PRODUCT=25	PASS
3	1	1000	SUM=1, PRODUCT=0	SUM=1, PRODUCT=0	PASS

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REQUIRED C PROGRAM: #include <stdio.h>

```
int main()
{
   int n, first, last, sum, product;
   int choice;

   printf("Enter a positive integer: ");
   scanf("%d", &n);

   if (n <= 0)
   {
      printf("Please enter a positive integer.\n");
      return 0;
   }

   printf("\n----- MENU -----\n");
   printf("1. Find sum and product of first and last digits\n");
   printf("2. Exit\n");

   printf("Enter your choice: ");
   scanf("%d", &choice);

   switch (choice)
   {
      case 1:
         first = n;
```

```
        last = n % 10;


        while (first >= 10)
        {
          first = first / 10;
        }


        sum = first + last;
        product = first * last;


        printf("Sum = %d\n", sum);
        printf("Product = %d\n", product);


        break;


      case 2:
        printf("Program terminated.\n");
        break;


      default:
        printf("Invalid choice.\n");
    }


    return 0;
}
```

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COMPILATION AND EXECUTION

```
┌──(kali㉿kali)-[~]
└─$ cc product.c -o product

┌──(kali㉿kali)-[~]
└─$ ./product
Enter a positive integer: 12345

———MENU———
1. Find sum and product of first and last digits
2. Exit
Enter yor choice: 1
Sum = 6
Product = 5

┌──(kali㉿kali)-[~]
└─$ ./product
Enter a positive integer: 5

———MENU———
1. Find sum and product of first and last digits
2. Exit
Enter yor choice: 1
Sum = 10
Product = 25
```

```
┌──(kali㉿kali)-[~]
└─$ ./product
Enter a positive integer: 1000

———MENU———
1. Find sum and product of first and last digits
2. Exit
Enter yor choice: 1
Sum = 1
Product = 0

┌──(kali㉿kali)-[~]
└─$ 
```

....